

DUTCHBAY TECHNICAL ADVISORY

Detailed Tender Gap Statement

Envision Energy 10 MW / 40 MWh BESS design package against the
NSO 250 MW / 1000 MWh standalone BESS procurement

Tender TR/REP&PM/ICB/2026/001/C

Prepared for Envision Energy via Hayleys

Review date: 31 July 2026

Report version: 1.0

Reviewed OEM document: Design Calculation Description for Sri Lanka 10 MW/40 MWh project, V1.0, 29 July 2026

Tender submission deadline in reviewed advertisement: 14 August 2026 at 10:00 Sri Lanka time

Confidential - authorized tender team use only

Important notice and review boundary

This report contains critique and derived data from an Envision document marked confidential and privileged. It must not be published in a public repository, website, data room, container image, or unrestricted tender library.

This is a technical and tender-readiness gap statement. It is intended to help Envision convert a short design calculation into a traceable, contract-aligned OEM tender package. It is not a legal opinion, an NSO acceptance decision, a certified grid study, or a substitute for the Project Proponent's complete feasibility study and proposal.

Sources reviewed

- NSO RFP Volume I: Instructions to Project Proponents, including Clauses 2.7, 2.8 and 3.1.
- NSO RFP Volume II: compliance schedule and Section 6 technical data forms.
- NSO RFP Volume III: model Energy Storage Agreement and payment/penalty provisions.
- NSO paper advertisement and the tender deadline.
- Envision design calculation V1.0 dated 29 July 2026.
- The supplied gap-review memorandum dated 30 July 2026.
- DutchBay repository evidence register, redacted ENPCS01 grid-code fixture, and grid-screening scope note.

Material source limitation

The reviewed Volume I refers to Annex A (Functional Performance Requirements), Annex B (Grid Connection Code), Annex C (embedded generator guide), and Annex D (SCADA signal list), but the bodies of those annexes are not present in the reviewed source files. Envision should obtain the complete issued annexes, every addendum and every clarification, then update the compliance matrix before tender freeze.

Interpretation discipline: where the attached review or repository narrative differs from the controlling RFP text, this report identifies the difference and recommends the more conservative compliance position until NSO clarifies it in writing.

Executive conclusion

The Envision calculation supports a plausible energy-sizing concept, but the reviewed OEM package is not tender-ready. It does not yet demonstrate the mandatory true grid-forming control mode, bankable dynamic performance, standards evidence, site-temperature guarantees, protection/safety design, or a reconciled AC architecture.

8

critical submission gates

10

major contractual or engineering gaps

4

useful design elements to retain

Immediate tender position

Dimension	Current position	Required position before submission
Technical responsiveness	Not demonstrated	One fixed configuration, completed Volume II forms, and clause-by-clause evidence tags.
Grid-forming compliance	Silent / adverse prior evidence	Signed true V/F declaration plus certified RMS and EMT evidence for the offered PCS/PPC release.
Energy and degradation	Numerically plausible, thin early margin	Guaranteed Termination/Metering Point values at site ambient with measurement tolerance and maintenance plan.
Round-trip efficiency	Year 15 below 85% in stated duty	Guaranteed monthly metered RTE schedule with margin, auxiliary boundary, remedies and back-to-back warranty.
Safety and standards	Evidence absent	Certificate register, type-test reports, UL 9540A fire propagation evidence and site protection design inputs.
Commercial bankability	Not addressed	OEM price validity, warranty, LTSA, spares and LD support aligned to flat-LKR capacity-charge exposure.

Recommended bid strategy: do not defend the ten-page calculation as the complete OEM tender response. Reissue it as a controlled Design Basis and Guaranteed Performance Schedule supported by a separately indexed compliance evidence dossier.

What should be retained

1. Coherent core sizing concept

The offered 44.2 MWh nameplate configuration reconciles arithmetically with the two stated DC blocks: 20.101 MWh plus 24.121 MWh equals 44.222 MWh. The stated BoL usable energy is 40.2 MWh at 33 kV.

2. Explicit 15-year performance table

The annual usable-capacity, system SoH, RTE and cumulative-export series is more decision-useful than a single end-of-life warranty point. It can become the basis of a controlled guarantee schedule if methodology and tolerances are added.

3. MDSC curve appears achievable

The deterministic auxiliary-inclusive curve stays above the NSO Minimum Dispatchable Storage Capacity schedule in all 15 years, from a 0.4 MWh margin in year 1 to 3.6 MWh in year 15.

4. Reactive target recognized

The calculation recognizes 10 MW and plus/minus 3.29 Mvar at 0.95 power factor at the PCC and attempts to translate that obligation back to the PCS terminals.

Do not overstate these positives

Positive	Why it is not yet a tender proof	Upgrade required
40.2 MWh BoL	Only 0.2 MWh, or 0.5%, above the requirement; the report itself allows 1% to 2% site/meter variation.	Guaranteed worst-case energy at +45 C, minimum DC voltage, auxiliaries online and metering tolerance.
15-year SoH curve	No cell model, temperature profile, throughput definition, warranty linkage or validation evidence.	OEM degradation model report and warranted annual capacity schedule.
RTE curve	No monthly/FEC test method, part-load profile, auxiliary boundary or uncertainty margin.	Metering-point guarantee and test protocol aligned with ESA Appendix A.
PQ chart	One chart at 0.9 pu does not demonstrate full operating-envelope or dynamic grid-forming performance.	Tabulated P-Q-V-SOC-temperature envelope plus RMS/EMT validation.

Critical submission gates - P0

P0-01 - True grid-forming V/F operation is not demonstrated

Requirement: Volume I Clause 3.1(c) requires a full grid-forming, voltage-sourced and voltage-controlled inverter that does not revert to current-controlled grid-following operation during normal operation or network faults.

Finding: the design calculation does not name the PCS model, PPC model, control release, grid-forming algorithm, mode-transition logic or fault-control behavior. The repository's earlier redacted ENPCS01 fixture is explicitly labelled GFL. That fixture is not proof about the offered 2026 product, but it is adverse prior evidence that must be resolved.

Required closure: signed OEM compliance declaration tied to exact hardware/firmware; control architecture; black-start/island/parallel-operation explanation; no-GFL-reversion statement; limitations register; model and test evidence.

P0-02 - PSS(R)E and PSCAD/EMTDC evidence is absent

Requirement: Clause 3.1(p) calls for initial non-site-specific PSS(R)E and PSCAD/EMTDC models, or specified test results, demonstrating V/P/Q response during deep and shallow faults and a +50-degree phase-angle step at SCR 1, 3, 5 and 10 with X/R 5.

Finding: the design calculation contains no model package, model guide, validation report, event plots or parameter checksum. Prior repository evidence refers only to PSS/E and DigSILENT GFL material; PSCAD is absent.

Required closure: provide both RMS and EMT model packages for the offered GFM release. Run both +50 and -50 degree steps to cover the controlling +50-degree text and the broader derived requirement. Include pass criteria, initialization cases and replay plots.

P0-03 - The AC architecture and ratings do not reconcile

The calculation states 15.1 MVA rated power and 14 MW usable active power at POC, but also states 4 x 2.52 MW PCS, a 10.1 MVA transformer and a 10.08 MW AC unit. Its own discharge-overexcitation case requires 11.24 MVA at PCS. The document does not show whether there are four or six PCS blocks, one or multiple AC skids, or which rating governs.

Required closure: one equipment schedule and SLD reconciling PCS count, kW/kVA ratings, overload duration, transformer windings, container mapping, losses, redundancy, auxiliary supply and the 10 MW export limit.

Critical submission gates - P0 continued

P0-04 - BoL energy margin is smaller than the stated uncertainty

The 40.2 MWh BoL value has only 0.2 MWh of headroom over the 40 MWh requirement. The calculation says actual conditions and metering can create 1% to 2% variation, equivalent to approximately 0.4 to 0.8 MWh. The stated uncertainty therefore exceeds the design margin.

The auxiliary case is calculated at 35 C although the site envelope extends to +45 C. Clause 3.1(k) requires four continuous hours at rated power under ambient site conditions at commissioning and says non-compliant offers will not be evaluated.

Required closure: increase guaranteed margin or prove a conservative worst-case calculation at +45 C and all other binding conditions. Separate expected, design and guaranteed values.

P0-05 - Monthly metered RTE guarantee has insufficient margin

The design curve falls to 84.9% including auxiliary power in year 15. The assumed 1.1 cycles/day is about 33 full equivalent cycles per month, which falls in the ESA band where the Guaranteed RTE is 85%. Even the 85.0% values in years 13 and 14 have no measurement or operating margin.

Volume I describes an 85% monthly minimum. Volume III applies 85%, 84%, 83% or 82% depending on monthly FEC. This inconsistency should be clarified, but the design should be priced and guaranteed on the conservative 85% case.

Required closure: guarantee metering-point RTE by FEC band, year, temperature, power and SOC window; define auxiliaries; include uncertainty; provide a test procedure and OEM remedy/LD support.

P0-06 - Mandatory standards evidence is not included

Clause 3.1(b) and Volume II Section 6 require documentary proof for extensive battery, PCS, safety, quality and grid standards. The calculation cites only EU Ecodesign for the transformer and does not include a certificate register or type-test mapping.

Required closure: controlled certificate matrix showing standard, edition, product/model, issuing laboratory, certificate/report number, validity, deviations and tender cross-reference.

Critical submission gates - P0 completed

P0-07 - Fire safety and thermal-runaway evidence is absent

Clause 3.1(g) requires fire prevention, detection, alarm, suppression, thermal-runaway mitigation, compartmentalization, emergency isolation and safety procedures, with evidence submitted with the proposal. The design calculation does not identify cell chemistry, UL 9540A results, propagation behavior, gas detection, ventilation, water strategy, fire zones or emergency response.

Required closure: product-level UL 9540/9540A and IEC evidence; hazard analysis; detection/suppression architecture; propagation and separation basis; emergency isolation philosophy; first-responder plan; responsibility matrix between Envision and EPC.

P0-08 - Plant SLD and export-limiting method are absent

Clause 3.1(e) requires a plant layout and SLD through the Grid Point, including switchgear, metering/protection CTs and VTs, transformers, inverters, and the method limiting maximum output to 10 MW +10% at the Termination Point.

Finding: no SLD, protection one-line, equipment boundary, metering point or limiter implementation is provided. This omission also prevents verification of auxiliary inclusion in RTE and resolution of the 10.08 MW / 14 MW / 15.1 MVA rating conflict.

Required closure: issue a tender SLD, protection SLD, communications architecture and export-limiter functional description with fail-safe behavior and test method.

Bid governance recommendation: none of P0-01 through P0-08 should be closed by narrative alone. Each requires an artifact, an owner, a document number, a revision, and an acceptance check in the master compliance schedule.

Design calculation critique - document control and design basis

Observed condition	Why it matters	Required improvement
Title, date and V1.0 only; no document number, status, author/checker/approver or revision history.	The evaluator cannot establish approval authority or trace later changes.	Add controlled title block, tender number, project/GSS, document ID, revision table and signatures.
Generic "Sri Lanka 10MW/40MWh" project with no selected GSS or site.	Ambient, fault level, route losses, auxiliary load and protection duties are site-dependent.	Issue a common OEM design basis plus a site schedule for each offered GSS.
No clause-by-clause requirement traceability.	Volume II warns missing information may cause rejection and requires item tags.	Add a compliance cross-reference beside every design input and output.
Conclusion reserves the right to amend data without prior notice.	That wording undermines reliance on tendered guarantees.	Separate non-binding indicative data from signed guaranteed particulars; freeze tender configuration.
Images are representative and scope may change.	The tender permits only one technical solution, not optional configurations.	Provide a final equipment list and state that alternatives require controlled approval.
BoP-owned assumptions include 100% MV/HV transformer and HV cable efficiency.	Zero-loss assumptions can overstate POC energy and RTE.	Use conservative guaranteed loss budgets and clearly allocate OEM/EPC interfaces.
No calculation method, software, model version or input provenance.	Performance values cannot be reproduced or independently checked.	Add calculation methodology, input files, cell model, solver/version, calibration and independent check record.
No explicit guarantee hierarchy.	Expected values can be mistaken for contractual guarantees.	Label every table value as requirement, expected, design, minimum guaranteed or informational.

Recommended replacement structure: (1) Design Basis, (2) Guaranteed Technical Particulars, (3) Performance Guarantee Schedule, (4) Calculations and simulations, (5) compliance evidence index, and (6) interface responsibility matrix.

Energy capacity, degradation and maintenance

Deterministic curve versus NSO minimum

Year	Envision usable MWh incl. aux	NSO minimum MWh	Deterministic buffer MWh	Finding
BoL	40.2	40.0	0.2	Below stated uncertainty
1	39.4	39.0	0.4	Tight
2	38.5	38.0	0.5	Tight
3	37.6	37.0	0.6	Tight
5	36.2	35.2	1.0	Positive deterministic buffer
10	33.2	31.2	2.0	Positive deterministic buffer
15	30.8	27.2	3.6	Positive deterministic buffer

Missing model definition

- Cell chemistry, cell model, initial dispersion and manufacturing tolerance.
- Calendar versus cycle degradation separation, C-rate, DoD, SOC window and temperature-time history.
- Definition of one FEC and consistency between 1.1 cycles/day and the 400 FEC/year obligation.
- Beginning-of-life clock: manufacture, shipment, FAT, SAT, COD and capacity-test date.
- Availability of spare modules/racks, balancing strategy and defective-cell replacement assumptions.
- Guaranteed measurement method at the Termination/Metering Point and tolerance treatment.

Capacity Maintenance Plan

A curve is not the Clause 3.1(m) plan. If Envision proposes no scheduled augmentation, the plan should say so expressly and show why, then define trigger-based contingency augmentation, reserved space, compatible future cell supply, outage durations, spares and recycling. Early-year margin is the binding issue, not year 15.

Round-trip efficiency and auxiliary consumption

Year	RTE incl. aux	RTE excl. aux	Auxiliary penalty	85% conservative test
BoL	86.3%	89.0%	2.7 pp	1.3 pp margin
5	85.6%	88.8%	3.2 pp	0.6 pp margin
10	85.2%	88.6%	3.4 pp	0.2 pp margin
13	85.0%	88.5%	3.5 pp	No margin
14	85.0%	88.5%	3.5 pp	No margin
15	84.9%	88.4%	3.5 pp	Fails by 0.1 pp

Why the current calculation is not guarantee-ready

- Auxiliary consumption is shown as 0.16 MW x 4.02 h at BoL but is explicitly said not to be separately guaranteed.
- No HVAC load curve is given for +45 C, humidity, standby, charge, discharge or fire-system states.
- No RTE result is given by power level, SOC window, FEC/month, voltage, aging state or grid-forming operating condition.
- No main-meter boundary diagram explains whether auxiliary import is included in charge energy.
- No PT/CT class, meter class, uncertainty budget or test stabilization procedure is provided.
- No reconciliation is made to ESA self-discharge adjustment rules and exclusions during Company maintenance/outage.

Commercial exposure: ESA liquidated damages value excess conversion losses at 150% of the applicable 33 kV General Purpose peak-time tariff. Envision's warranty and LTSA must support the Project Company for OEM-caused shortfall, including diagnostics and remediation.

PCS, transformer and reactive capability reconciliation

Statement in design calculation	Value	Reconciliation issue
Rated power	15.1 MVA	No equipment count or continuous/overload basis shown.
Usable active power at POC	14 MW	Exceeds the 10 MW contract point and is not connected to a documented limiter.
PCS rated power	4 x 2.52 MW	Totals 10.08 MW; unit is stated as MW while later PCS apparent-power requirement is in MVA.
Transformer / AC unit	10.1 MVA / 10.08 MW	Does not explain the separate 15.1 MVA system rating.
Worst stated PCS apparent-power requirement	11.24 MVA	Exceeds 10.08 if the four 2.52 units are each 2.52 MVA.
PCC target	10 MW, plus/minus 3.29 Mvar	Recognized, but only a static chart at 0.9 pu is presented.

Required reconciliation pack

1. Equipment schedule with exact part numbers, quantity, continuous rating, overload rating and derating curves.
2. SLD mapping every DC container to PCS, transformer winding, MV bus, auxiliary transformer and meter.
3. Four-quadrant P-Q capability at PCC for 0.9, 1.0 and 1.1 pu; charge/discharge; BoL/EoL; minimum/maximum SOC; and full site temperature range.
4. Transformer thermal loading and impedance study, including 9% impedance, harmonic loading and contingency operation.
5. Export-limiter design with PPC/PCS hierarchy, measurement source, setpoint range, response time, redundancy and fail-safe action.
6. Loss allocation from DC terminals to the contractual metering point with worst-case tolerances.

Do not submit both 10.08 MW and 14 MW as unexplained "available" ratings. The RFP permits one technical solution. Extra installed power can be valuable for reactive headroom and aging, but it must be described as intentional oversizing governed by a tested 10 MW export cap.

Grid-forming and dynamic model acceptance specification

Deliverable	Minimum content	Acceptance check
GFM compliance statement	Exact PCS/PPC hardware and firmware; voltage-source/voltage-control declaration; no GFL reversion in normal/fault operation; operating limitations.	Signed by authorized Envision engineering representative and cross-referenced to models.
Control philosophy	Voltage/frequency source, droop, virtual impedance/inertia if used, current limiting, fault logic, setpoint hierarchy, parallel operation.	No undisclosed PLL/current-control fallback; stable mode logic across use cases.
PSS(R)E package	Model files, libraries, parameter files, model guide, initialization case, source code or escrow terms where required, checksum.	Loads without error; initializes; reproduces validation plots and stated event response.
PSCAD/EMTDC package	EMT project, compiled components, model guide, time-step guidance, initialization, parameter mapping, checksum.	Stable at recommended time step and reproduces V/P/Q response.
Validation report	Deep/shallow faults; SCR 1/3/5/10; X/R 5; +50 degree step and recommended -50 degree companion; charge/discharge; multiple SOC points.	Pass/fail criteria stated before results; plots include time, voltage, P, Q, current and controller states.
Model governance	Release notes, version compatibility, parameter-change process, confidentiality/licensing and support contacts.	Tender model matches supplied hardware/firmware and later site model.

Critical distinction

The DutchBay in-house grid screens are advisory RMS/design checks. They cannot certify OEM control-loop behavior or replace PSCAD/EMTDC. Likewise, the redacted ENPCS01 fixture carries useful prior parameters, but it is explicitly non-executable and grid-following. The tender response must not cite either as proof of the offered GFM product.

Schedule protection

Volume I also requires Dynamic Model Test results within two months of accepting the Letter of Award and bars construction until NSO concurrence. Envision should commit named model engineers, turnaround times, license support and a pre-submission model-validation workshop so this does not become a notice-to-proceed critical path.

Safety, standards and protection evidence

Battery and system standards

- IEC 62619, IEC 62620, IEC 62902.
- IEC 62485-5 and IEC 63056.
- IEC 62933-1, IEC 62933-2-1 and IEC 62933-5-2.
- UL 9540, UL 9540A and UL 1973.
- UN 38.3 transport evidence.
- ISO 9001:2015 and ISO 14001:2015.

PCS/grid standards

- IEEE 1547-2018 and IEEE 2800-2022.
- UL 1741-SB.
- IEC 62477-1 and IEC 62109-1/2.
- IEC TS 62786-3.
- IEC 61000-6-2/6-4 and IEC 62116.

Protection and site inputs

- 33 kV GSS maximum three-phase fault duty ranges from 4.5 kA at Monaragala to 16.2 kA at Kiribathkumbura in the reviewed schedule.
- Provide PCS fault-current contribution, duration and control behavior.
- Provide AC/DC overcurrent, short-circuit, overvoltage, temperature, ground-fault and reverse-polarity protection.
- Provide transformer, busbar, feeder and inverter relay functions, settings philosophy and coordination inputs.
- Provide earthing, lightning, surge, arc-flash and emergency-isolation design data.

Evidence quality

A brochure statement is not documentary proof. Each certificate must match the exact tendered model or show an engineering applicability bridge approved by the issuing body.

UL 9540A is not a single generic certificate. Envision should state the tested cell/module/unit/installation level, test configuration, propagation result and design deltas between the test article and offered container.

Major engineering and operational gaps - P1

ID	Gap	Required Envision deliverable	Acceptance condition
P1-09	Capacity Maintenance Plan absent	Initial sizing, scheduled/triggered augmentation, replacement intervals, outage interface, recycling and decommissioning.	Shows annual compliance under guaranteed downside, not expected curve only.
P1-10	Availability basis absent	Reliability block diagram, failure rates, redundancy, MTTR, planned maintenance, spares and service coverage.	Supports 97% monthly availability and annual reconciliation.
P1-11	Service-life evidence absent	20-year non-battery and 15-year cell/module/rack design-life declarations plus warranty schedule.	Duty cycle and environmental envelope match ESA/tender.
P1-12	Thermal management not defined	HVAC architecture, setpoints, redundancy, parasitic curves, humidity/corrosion limits and derating.	Energy, RTE and life guarantees proven at site conditions.
P1-13	EMS/PPC/SCADA package absent	PPC model, EMS functions, NSCC communications, backup channel, signal list, event records and cybersecurity.	Supports dispatch, GFM setpoints and monthly A.05 evidence reports.
P1-14	Metering/monitoring absent	Main meter boundary, grid-code compliance meter, PQ/FRT monitoring, data retention and report formats.	Monthly energy, RTE, availability and SoH are auditable.
P1-15	OEM qualification evidence absent	Certified sales records for 1 GWh battery cells/modules, 1 GW inverter products and two same-family PPC projects in last three years.	Meets Clause 2.7.3 with manufacturer certification.
P1-16	Noise and environmental details absent	Noise at boundary/equipment, hazardous materials, water, waste, recycling and transport data.	Completes Volume II fields and site permitting inputs.
P1-17	FAT/SAT/capacity test undefined	Inspection and test plan with hold points, calibrated instruments, acceptance criteria and retest/remedy process.	Contractual energy/RTE/PQ/GFM guarantees are objectively testable.
P1-18	Interface allocation incomplete	OEM/EPC/BoP/Project Company/NSO responsibility matrix.	No orphaned guarantees for cables, transformer, auxiliaries, controls or models.

Commercial and contractual alignment for Envision

Project Company exposure that should be backed by the OEM

ESA/RFP exposure	Risk to Project Company	Recommended OEM support
Flat LKR capacity charge for 15 years; one-time 85% USD adjustment only at ESA signing	Long-term USD/CNY equipment and service cost inflation is not automatically passed through.	Price validity through award/ESA, defined FX basis, capped escalation for LTSA/spares and transparent local/foreign split.
97% monthly availability with double-capacity-charge LD mechanics and annual reconciliation	Equipment failures or slow OEM response directly reduce revenue.	Back-to-back availability warranty, response times, spares, remote diagnostics and OEM-caused LD indemnity.
Annual ADSC versus MDSC capacity-charge reduction	Capacity underperformance reduces the subsequent year's monthly revenue.	Annual capacity warranty, test protocol, augmentation/remedy obligation and cost allocation.
RTE liquidated damages	Metered conversion losses are valued at 150% of peak tariff.	RTE guarantee at contractual meter, correction plan, test tolerances and back-to-back loss compensation.
Functional-performance LD: 1% of monthly capacity charge per event, capped at 10%	Control/model non-performance creates repeated event exposure.	GFM/grid-support warranty, firmware support, root-cause SLA and OEM-caused LD support.
Total monthly LD cap of 20% of capacity charge	Multiple performance failures can stack.	Aggregate OEM liability carve-out adequate for performance-caused losses, without excluding model/control failures.

Minimum commercial annexes requested from Envision

- Binding scope and price schedule with Incoterms, taxes, currency, validity and delivery milestones.
- Guaranteed particulars and warranty schedule aligned to the final technical configuration.
- LTSA term sheet covering preventive/corrective maintenance, remote support, spares and software.
- Performance security and parent-company support position.
- Back-to-back availability, capacity, RTE and functional-performance remedies.
- Technology obsolescence, augmentation compatibility and cybersecurity/firmware support commitments.

Required OEM submission dossier

Volume	Required documents	Owner	Freeze gate
O-01 Document control	OEM cover letter; master document register; compliance matrix; deviation register; confidentiality schedule.	Tender manager	No blank or untagged requirement.
O-02 Design basis	Site envelope; duty cycle; grid data; codes; interfaces; design life; design margins.	Lead engineer	Inputs approved by EPC/ Project Company.
O-03 Configuration	Equipment schedule; SLD; layout; container/rack/PCS mapping; auxiliary architecture; export limiter.	System engineering	All ratings reconcile.
O-04 Performance	Energy, power, RTE, degradation, PQ, availability and auxiliary guarantees with test conditions.	Performance engineering	Worst-case margins positive.
O-05 GFM controls	GFM declaration; control philosophy; PPC/EMS description; droop/deadband/setpoint ranges.	Controls engineering	No GFL fallback ambiguity.
O-06 Dynamic models	PSS(R)E; PSCAD/EMTDC; model guides; validation; checksums; support plan.	Grid integration	Independent replay passed.
O-07 Electrical/ protection	Transformer/switchgear data; fault contribution; protection; earthing; harmonics/flicker.	Electrical engineering	Selected GSS duty confirmed.
O-08 Safety/ environment	Fire package; UL 9540A applicability; HAZID/FMEA; emergency response; noise; recycling.	HSE/product safety	Evidence maps to offered product.
O-09 Standards	Certificate/type-test register and full reports.	Quality	No expired or wrong-model evidence.
O-10 Operations	Capacity Maintenance Plan; O&M plan; spares; training; monthly reporting; cybersecurity.	Service organization	15-year support demonstrated.
O-11 Qualification	Certified installed base and same-family PPC references.	Commercial/quality	Clause 2.7.3 thresholds met.
O-12 Commercial	Binding offer; warranty; LTSA; remedies; schedule; securities.	Commercial/legal	Back-to-back review completed.

Packaging principle: every row in the Project Proponent's compliance schedule should cite one or more exact Envision document numbers and page/section references. "Comply" without an evidence pointer should be rejected during internal red-team review.

Recommended clarification questions to NSO

1. **RTE hierarchy:** confirm whether the controlling guarantee is the 85% monthly minimum in Volume I or the FEC-banded 85/84/83/82% schedule in Volume III, and confirm the governing order if inconsistent.
2. **RTE metering boundary:** confirm treatment of auxiliaries supplied behind or separately from the main meter, standby consumption, HVAC, fire systems and transformer losses.
3. **Dynamic evidence route:** confirm whether test results can substitute for both PSS(R)E and PSCAD models at tender stage, or whether both models are expected regardless; Envision should plan to provide both.
4. **Phase-angle test:** controlling Clause 3.1(p) states +50 degrees, while derived tender notes refer to plus/minus 50 degrees. Confirm whether both polarities are required.
5. **GFM current limiting:** confirm required fault current magnitude/duration and acceptable voltage-source current-limiting behavior without classification as GFL mode.
6. **Black start/island operation:** confirm whether these are required capabilities or only true grid-forming parallel operation.
7. **PQ envelope:** confirm required P-Q capability across 0.9 to 1.1 pu, SOC, temperature, charge/discharge and EoL conditions.
8. **Capacity test timing:** confirm whether BoL and annual ADSC tests are at Termination Point or Metering Point, and the test SOC/temperature/auxiliary conditions.
9. **Annex package:** provide the complete issued Annex A-D, addenda and official clarification log if not already supplied.
10. **Export limit:** confirm whether 10 MW +10% is an instantaneous ceiling, a test tolerance or the maximum limiter setting, and define measurement/response time.

Do not wait for clarifications to start design closure. Engineer the conservative case now, then relax only where an NSO written response clearly permits it.

Two-week closure plan to tender freeze

Timing	Envision action	Output	Decision gate
Day 0-1	Appoint one technical configuration owner; freeze exact battery, PCS, PPC, transformer and firmware.	Configuration baseline and master document register.	No alternative solution remains in bid.
Day 0-2	Reconcile 10.08 MW, 11.24 MVA, 14 MW and 15.1 MVA statements; issue SLD and loss diagram.	Signed rating reconciliation note.	All quantities and ratings close.
Day 1-3	Issue true-GFM declaration and control philosophy; identify model release and responsible engineers.	GFM dossier draft.	No GFL fallback ambiguity.
Day 1-4	Recalculate energy/RTE at +45 C and contractual meter with conservative tolerances.	Guaranteed performance schedule.	Positive margin at all guarantee points.
Day 2-5	Assemble standards, fire, qualification and design-life evidence.	Evidence register with full attachments.	Every Volume II field supported.
Day 2-7	Run/replay RMS and EMT events and prepare validation report.	PSS(R)E/PSCAD package and plots.	SCR/event test matrix passed.
Day 4-8	Complete Capacity Maintenance, availability, O&M, spares and monthly reporting plans.	15-year service dossier.	Operational obligations resourced.
Day 6-9	Align warranty/LTSA/remedies to Project Company ESA exposure.	Commercial annexes.	Back-to-back gap register accepted.
Day 9-11	Populate Volume II forms and compliance schedule with evidence tags.	OEM submission pack integrated into bid.	No blank/unsupported answer.
Day 11-12	Independent red-team review and document freeze.	Closed punch list and signed release.	Submission authorization.

The sequence is deliberately parallel. If validated GFM EMT evidence does not already exist, this item is likely to dominate the schedule and must be escalated immediately.

Detailed compliance and evidence matrix

ID	RFP reference	Requirement	Current Envision evidence	Gap rating	Required evidence / acceptance
C-01	Vol I 3.1(a); Vol II Sec. 6	Complete technical particulars	Partial sizing and PCC tables only	Critical	Complete every A-E field with exact model data and attachments.
C-02	Vol I 3.1(b)	Battery and PCS standards with proof	No certificates	Critical	Certificate/type-test matrix tied to exact product.
C-03	Vol I 3.1(c)	True GFM V/F, no GFL reversion	Silent; prior fixture GFL	Critical	Signed declaration, control description and model/test proof.
C-04	Vol I 3.1(d)	Transformer and switchgear details	Voltage, MVA and impedance only	Major	Datasheets, ratings, derating, losses, short-circuit duty and standards.
C-05	Vol I 3.1(e)	Layout, SLD and 10 MW +10% limiter	Absent	Critical	Controlled drawings and limiter functional/test description.
C-06	Vol I 3.1(f)	OEM expression of interest and capability	Not in calculation	Major	Signed supply EOI and corporate/product literature.
C-07	Vol I 3.1(g)	Fire prevention/detection/suppression	Absent	Critical	Fire dossier, test applicability, mitigation and emergency response.
C-08	Vol I 3.1(h)	Equipment ratings and protection at GSS fault level	Impedance values only	Major	Fault withstand, contribution and coordinated protection inputs.
C-09	Vol I 3.1(i)	Metering, recording and NSO transmission	Absent	Major	Meter/PQ monitor/communications/data architecture.
C-10	Vol I 3.1(k)	Four hours at rated power under commissioning ambient	40.2 MWh at 35 C auxiliary basis	Critical	Worst-case +45 C guarantee and acceptance test margin.
C-11	Vol I 3.1(l)	20-year non-battery / 15-year battery life	15-year curve, no declaration	Major	Design-life and warranty declarations tied to duty.
C-12	Vol I 3.1(m)	Capacity Maintenance Plan	Curve only	Major	Initial sizing, augmentation, replacement, outage and recycling plan.
C-13	Vol I 3.1(n)	Monthly energy/RTE/availability/SoH reporting	Absent	Major	EMS data dictionary, retention and sample report.
C-14	Vol I 3.1(o)	A.05 grid support and EMS event evidence	Static PCC table only	Critical	Full Annex A matrix, controls evidence and event-report design.
C-15	Vol I 3.1(p)	PSS(R)E + PSCAD or specified test evidence	Absent	Critical	Executable models, guides, event matrix and validation report.
C-16	Vol I 2.8.1; ESA App. A	97% monthly availability	No reliability analysis	Major	RBD, maintenance, spares and warranty support.
C-17	Vol I 2.8.1; ESA App. A	MDSC annual compliance	Expected curve above floor	Major	Guaranteed downside curve and maintenance triggers.
C-18	Vol I 2.8.1; ESA App. A	Monthly metered RTE	84.9% year 15 incl. aux	Critical	85% conservative guarantee with margin and test method.
C-19	Vol I 2.7.3	OEM installed-base qualification	Absent	Major	Manufacturer-certified 1 GWh, 1 GW and two PPC project records.
C-20	Vol II B.10-B.40	Harmonics, droop, deadband, LVRT and GFM	PF/PQ only	Critical	Completed forms plus datasheets and validation evidence.

Internal red-team acceptance checklist

No.	Question before OEM pack release	Pass evidence	Owner	Status
1	Does one frozen configuration reconcile all DC, PCS, transformer and POC ratings?	Signed equipment schedule + SLD + rating note	System lead	Open
2	Is true GFM behavior explicitly guaranteed without GFL fault fallback?	Signed declaration + control philosophy	Controls lead	Open
3	Do PSS(R)E and PSCAD models match the tendered firmware?	Model release record and checksum	Grid lead	Open
4	Have all SCR/fault/phase-step cases been replayed independently?	Validation matrix with pass criteria	Independent checker	Open
5	Is 40 MWh guaranteed at +45 C with auxiliaries and meter uncertainty?	Worst-case calculation and FAT/SAT procedure	Performance lead	Open
6	Is monthly metered RTE guaranteed with positive margin in every year?	RTE schedule and test protocol	Performance lead	Open
7	Does the Capacity Maintenance Plan close early-year uncertainty as well as EoL?	Downside capacity model and triggers	Service lead	Open
8	Do certificates match exact offered models and editions?	Quality-approved certificate register	Quality lead	Open
9	Does UL 9540A evidence map to offered container/rack configuration?	Applicability statement and design-delta review	Safety lead	Open
10	Are all site fault levels within equipment duty and protection study inputs?	GSS schedule and signed ratings check	Electrical lead	Open
11	Can EMS generate every monthly NSO report and event record?	Signal/data/report mapping	EMS lead	Open
12	Does the OEM support Project Company LD exposure?	Commercial back-to-back matrix	Commercial lead	Open
13	Are all Volume II technical fields completed and tagged?	Form completion audit	Tender manager	Open
14	Are Annex A-D, addenda and clarifications incorporated?	Document register and compliance delta review	Document control	Open
15	Has an independent person attempted to disprove every claimed compliance?	Red-team punch list closed	Bid director	Open

Technical release

Name / signature / date:

Commercial release

Name / signature / date:

Bid director release

Name / signature / date:

Source register and clause notes

Source	Primary use in this report	Control note
NSO RFP Volume I, July 2026	Submission requirements, technical Clauses 3.1(a)-(p), commercial terms, evaluation and schedule.	Controlling procurement instruction within reviewed files.
NSO RFP Volume II, July 2026	Compliance schedule and Section 6 technical data fields.	Blank fields must be completed and evidence-tagged.
NSO RFP Volume III, model ESA	Availability, MDSC, RTE and functional-performance payment/LD mechanics.	Contract form deemed accepted on proposal submission under Volume I.
NSO paper advertisement	Procurement scope and deadline.	Deadline stated as 14 August 2026 at 10:00.
Envision design calculation V1.0, 29 July 2026	OEM configuration, capacity, degradation, RTE, assumptions and PCC data.	Confidential; no public distribution.
Gap review dated 30 July 2026	Initial issue identification.	Derived review; corrected/expanded against controlling RFP text.
DutchBay ENPCS01 redacted fixture	Prior GFL topology and reference parameters.	Adverse prior evidence only; not executable and not proof about offered 2026 product.
DutchBay grid-screening scope note	RMS versus EMT boundary and non-bankable limitation.	Internal advisory methodology, not tender compliance evidence.

Corrections and refinements to the supplied short gap review

- Controlling Clause 3.1(p) in the reviewed Volume I states a **+50-degree** phase-angle step. The supplied review and internal note say plus/minus 50 degrees. This report recommends testing both polarities while recording +50 degrees as the minimum explicit text.
- Volume III uses FEC-banded Guaranteed RTE thresholds, while Volume I presents an 85% monthly minimum. This is a tender inconsistency requiring clarification; Envision's 1.1 cycles/day assumption nevertheless points to the 85% band.
- The critical problem is not only that BoL energy was calculated at 35 C. The 0.5% energy margin is itself smaller than the calculation's stated 1% to 2% variability.
- The apparent-power and equipment-count mismatch is a separate critical gap because it can invalidate both the PQ claim and the SLD/export-limiter design.
- The current source set does not include the bodies of Annex A-D, so no final grid-support compliance claim should be made until those issued documents are reviewed.

Prepared through the DutchBay WeasyPrint presentation layer. Numerical values are transcribed from the reviewed documents; no OEM warranty or NSO acceptance is inferred.